

Volume I · Chapter 0 — Python & GitHub Fundamentals

· Labs

Onboarding chapter (added after V1), maintained independently. Hour model **1 : 2 : 4** (theory : labs : homework). These are **supervised, hands-on** exercises reinforcing the Theory topics. See `Volume-I-Chapter-0-context.md`.

Prerequisites: Chapter 0 Theory (0.1–0.11).

Learning outcomes (labs): the student can independently set up the toolchain, write and run Python programs covering every core construct, and perform the full Git/GitHub workflow (commit → branch → merge/conflict → push → pull request).

Labs (hands-on) — 20 h:

Python (14 h)

#	Lab	h
0a	Install Python + toolchain; run first scripts from the terminal and in Jupyter	2
0b	Variables, types & string manipulation exercises	2
0c	Control-flow drills: conditionals, loops, break/continue	2
0d	Data structures: build & manipulate lists / tuples / dicts / sets (+ comprehensions)	2
0e	Write functions; organize code into modules; install & use a package with pip	2
0f	File I/O + exception-handling mini-exercises	2
0g	Mini-project: a small CLI script (e.g., a calculator or a text/CSV processor)	2

Git & GitHub (6 h)

#	Lab	h
0h	Install & configure Git; init a repo; stage & commit; write a .gitignore	2
0i	Branching & merging; deliberately create a merge conflict and resolve it	2
0j	GitHub <i>Hello World</i> : create a repo, clone, push, open and merge a Pull Request	2

Datasets/tools: Python 3, VS Code/Jupyter, pip, Git, a GitHub account; a sample text/CSV file for Lab 0f/0g.

Assessment (labs): lab rubric across 0a–0j (completion + correctness + clean commit history). The Lab 0g mini-project and the Lab 0j merged PR are the key deliverables.

Key decisions: terminal vs. notebook for a task; how to structure a small script into functions/modules; commit granularity and branch naming.

References: see the Theory file's "References & learning resources" (W3Schools Python/Git, Python.org tutorial, GitHub Hello World & learning resources).

Hours: Labs **20** (paired with Theory 10 and Homework 40; ratio 1 : 2 : 4).